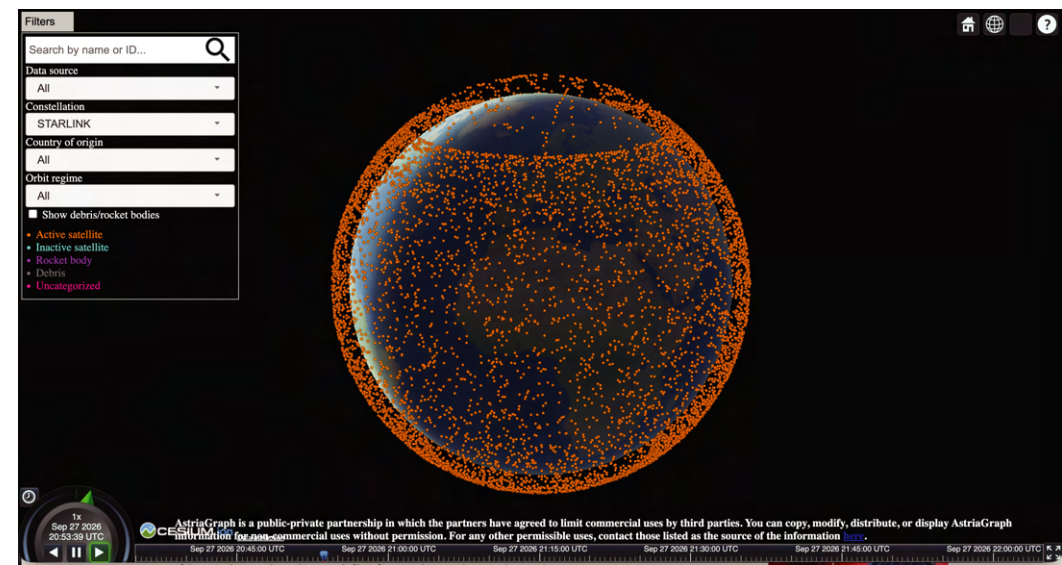
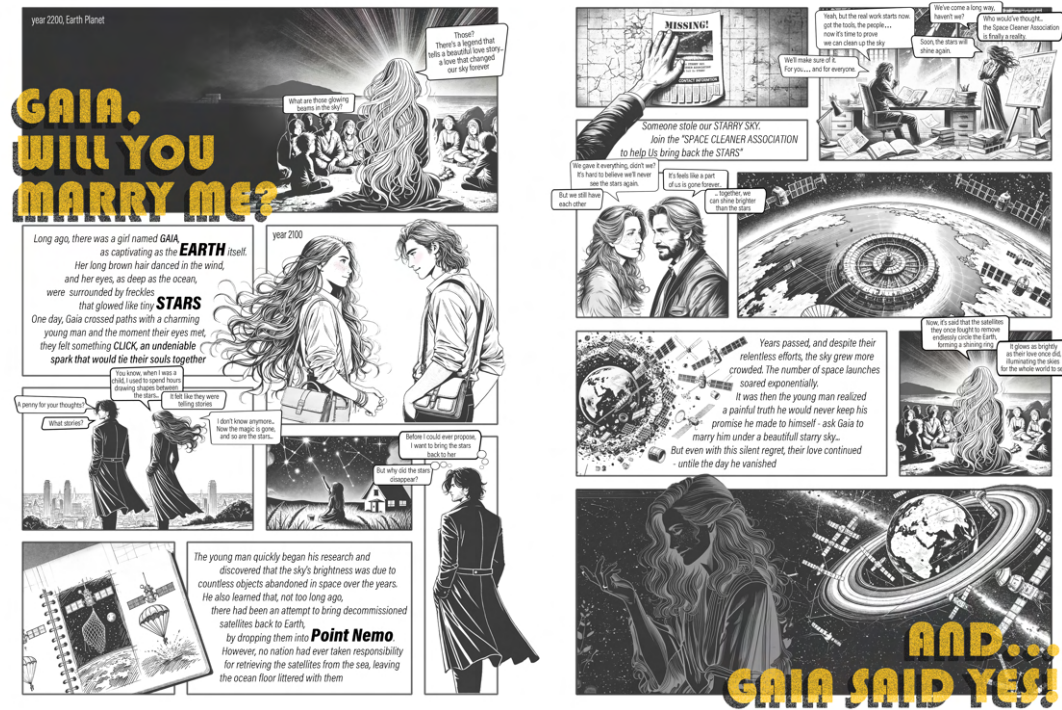


+ CLAIMING THE SKY



<http://astria.tacc.utexas.edu/AstriaGraph/>

#01 WHAT AM I ATTEMPTING?

I want to explore the future of Starlink satellites used for internet connectivity in low Earth orbit (LEO). Starting from their launch dates and physical characteristics, I aim to estimate when they might reenter the atmosphere and how much aluminum oxide they could produce. The model would estimate oxide production; potential effects on ozone motivate the research but would not be calculated directly.

#02 THE DATASET

I will use the General Catalog of Artificial Space Objects (GCAT), compiled by Jonathan C. McDowell, first published in 2020 and regularly updated. It documents artificial space objects, including satellites, rocket stages, components, and debris.

The main file, satcat.tsv, displays 70,001 rows and 42 columns in Numbers. This includes introductory rows and objects other than satellites.

The columns I will use are:
[Name] [LDate] [DDate] [Status] [DryMass in Kg] [Owner] [Perigee] [Apogee]

#03 RESESARCH

Official Starlink documentation states that satellites are designed to serve for five or more years after reaching their operational orbits. Retirement decisions depend on each satellite's condition. I will use this as a reference for an assumed service life, rather than a fixed retirement deadline.

Starlink link:
https://space-safety.starlink.com/docs/space-safety-articles/constellation_altitudes/

For oxide production, I will use Ferreira et al. (2024), Potential Ozone Depletion From Satellite Demise During Atmospheric Reentry in the Era of Mega-Constellations. Section 3.3 estimates approximately 29.8 kg of aluminum oxide clusters from a 250 kg satellite containing 30% aluminum. Applying this result to Starlink satellites will be an explicit modeling assumption.
Research paper link: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL109280>

#04 MY RECORD

Each starting row will represent one Starlink satellite still in orbit

For each satellite, I will retain its identifier, launch date, dry mass, and available orbital characteristics. I will add model fields for age, projected retirement date, projected reentry date, and estimated oxide production.

#05 MY RULE

At each annual step, every satellite becomes one year older.
The proposed sequence is

I will use service lives of 5 years. These are scenarios I have chosen, informed by Starlink's stated design life of "5+ years." Using the launch date as the beginning of service neglects the time required to reach the operational orbit.

For an initial annual model, I will assume reentry occurs in the same year as

retirement. Satellites still in orbit but already older than the assumed service life will be flagged for reassessment rather than assigned a fictional past reentry.

For each satellite projected to reenter:

oxides produced (kg) = reentry mass (kg) × aluminum mass fraction (30% of the mass) × production coefficient (k).

I will use DryMass as an approximation of reentry mass. In the reference scenario, the aluminum fraction will be 0.30, and the production coefficient will be:

$k = 29.8 / (250 \times 0.30) \approx 0.397$ kg of oxides per kg of initial aluminum.

These values come from the case modeled by Ferreira et al. Their applicability to different Starlink generations is a major uncertainty. The output refers to aluminum oxide clusters, not exclusively pure Al_2O_3

#06 MINE RUN

The model will move forward in annual steps for ten years from the reference date. It will track only satellites in the starting record; future launches could be added in a later extension.

The first version will be deterministic. Different combinations of service life and reentry delay will show how sensitive the results are to these assumptions.

#07 THE DATA IT WOULD GENERATE

In my visualization, the horizontal axis will represent time and the vertical axis altitude. Each satellite will appear as a band between its perigee and apogee, extending schematically toward its projected reentry. At reentry, a circle's area will represent the estimated mass of oxides produced. The bands will not be physical trajectories, and the circles will not represent atmospheric dispersion. A companion bar chart will show annual oxide production in tonnes.

#08 WHAT IT CAN'T SEE

- < Actual retirement decisions, early failures, and future operational changes. >
- < Detailed orbital evolution and the effects of solar activity. >
- < Each satellite's exact composition and differences in reentry chemistry. >
- < Future launches, collisions, and fragmentation. >
- < Atmospheric transport, concentration, and persistence of the oxides. >
- < Quantitative ozone depletion or other environmental damage. >

